Introduction to GitHub

Creating your first blog



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GitHub Red Hat

What we'll discuss today

- **GitHub**

  Quick overview of what GitHub is

- **Pull Requests**

  What Pull Requests are and why we use them

- **Hands-on exercise**

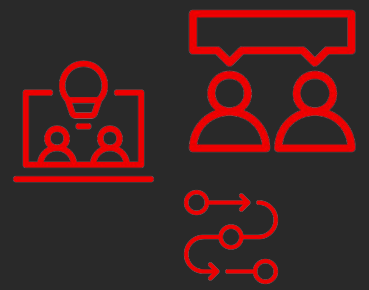
  Write your own blog post and add it to our TY website

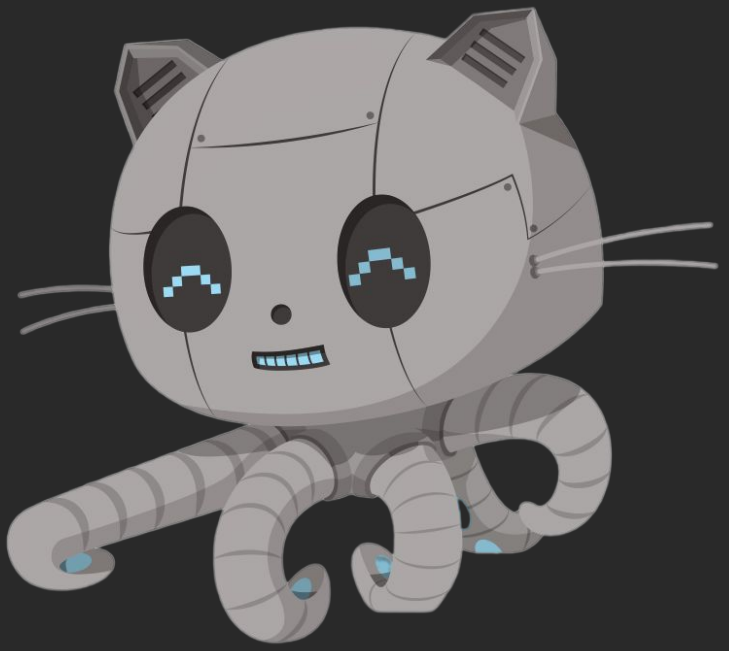




GitHub

What is GitHub and Why is it important?





Evie and Grettel work on a project together from different computers



- Both developers need to apply changes to the same project.
- They need access to the latest version of the project (others might be also collaborating).
- Evie and Grettel have different tasks.
- They must work simultaneously without affecting the main project.

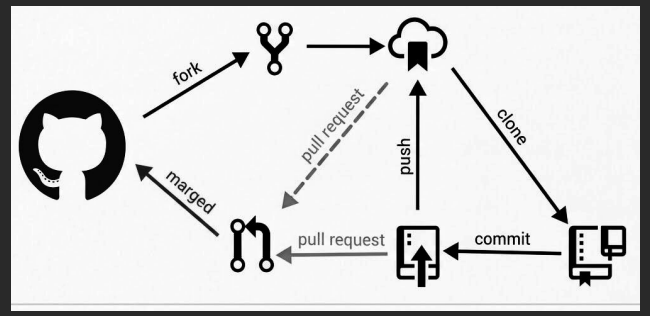
So how do Evie and Grettel solve these challenges?

They need a tool that lets them:

- Work on the same project without getting in each other's way.
- Always have access to the latest version.
- Work on different tasks at the same time.
- Keep the main project safe while they experiment.
- **This is exactly what GitHub does!**







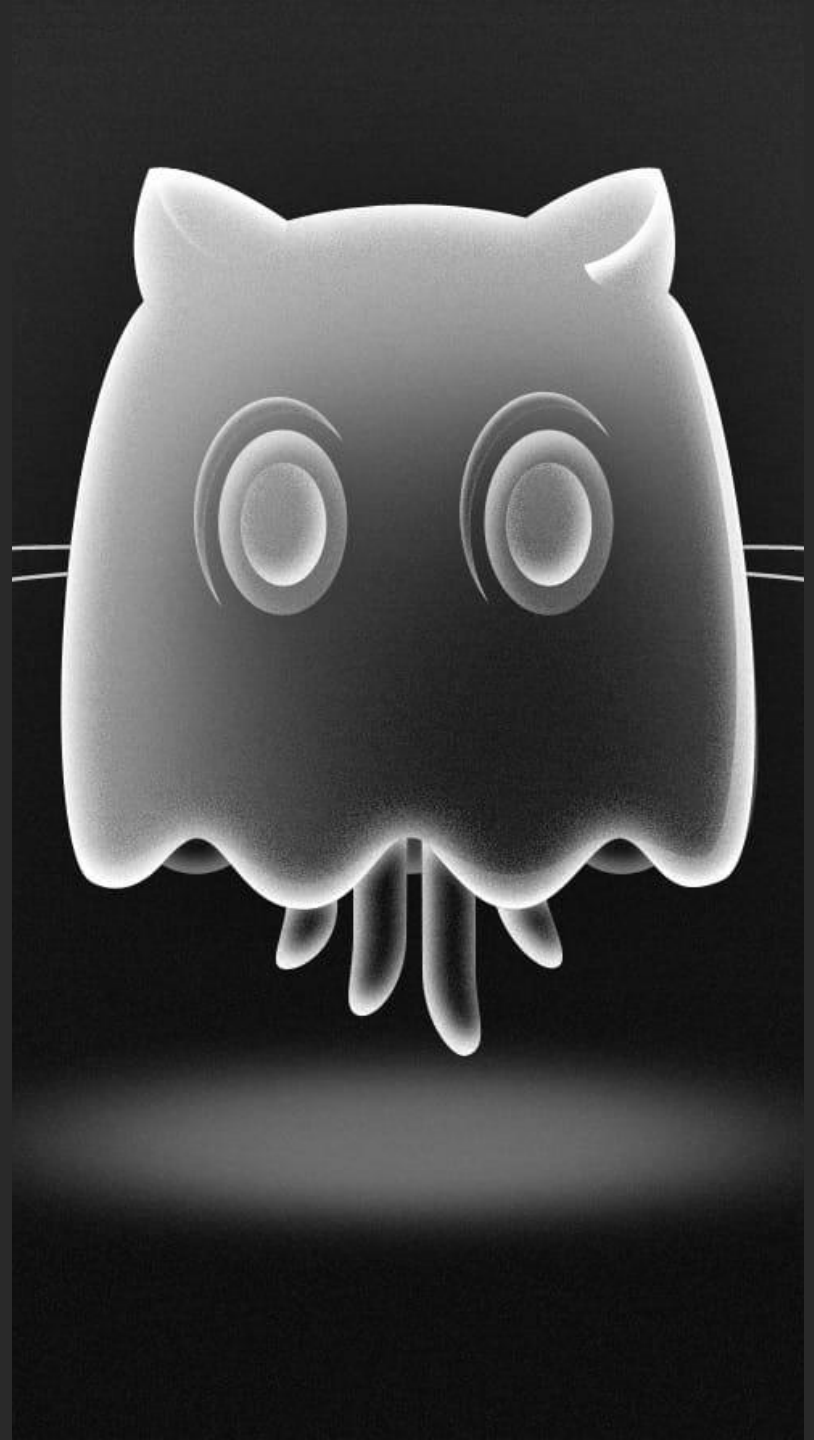
What is GitHub?

GitHub is a platform for building software. Whether you are starting a new project or contributing to an existing one, GitHub brings together the tools and workflows you need.

GitHub is based on the open-source software, **Git.**

Git is a **version control system** that tracks changes to files. It lets you work on your own copy of a project, record a history of your changes, and merge your work with other people's changes safely.

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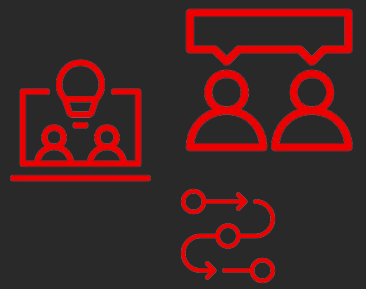
Why is GitHub important?

GitHub is essential because it makes software development **organized, collaborative, and safe** - like upgrading from writing essays on napkins to having a professional writing studio with backups, editors, and publishing tools.

For students learning to code, GitHub teaches you how real developers work in companies. It automatically saves every version of your work so **you never lose progress and it lets you safely experiment with new ideas** without breaking what already works.

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Pull Requests



How do we make sure our code stays safe?

```
30  31    <!-- Sidebar/menu -->
31      - <nav class="w3-sidebar w3-red w3-collapse w3-top w3-large w3-padding" style="z-index:3;width:300px;font-weight:bold;" id="mySidebar"><br>
        32 + <nav class="w3-sidebar w3-blue w3-collapse w3-top w3-large w3-padding" style="z-index:3;width:300px;font-weight:bold;" id="mySidebar"><br>
32  33      <a href="javascript:void(0)" onclick="w3_close()" class="w3-button w3-hide-large w3-display-topleft" style="width:100%;font-size:22px">Close Menu</a>
33  34      <div class="w3-container">
34  35        <h3 class="w3-padding-64"><b>Company<br>Name</b></h3>
        @@ -44,8 +45,8 @@ <h3 class="w3-padding-64"><b>Company<br>Name</b></h3>
44  45    </nav>
45  46
46  47    <!-- Top menu on small screens -->
47      - <header class="w3-container w3-top w3-hide-large w3-red w3-xlarge w3-padding">
48      -   <a href="javascript:void(0)" class="w3-button w3-red w3-margin-right" onclick="w3_open()">≡</a>
        48 + <header class="w3-container w3-top w3-hide-large w3-blue w3-xlarge w3-padding">
        49 +   <a href="javascript:void(0)" class="w3-button w3-blue w3-margin-right" onclick="w3_open()">≡</a>
49  50      <span>Company Name</span>
50  51    </header>
```





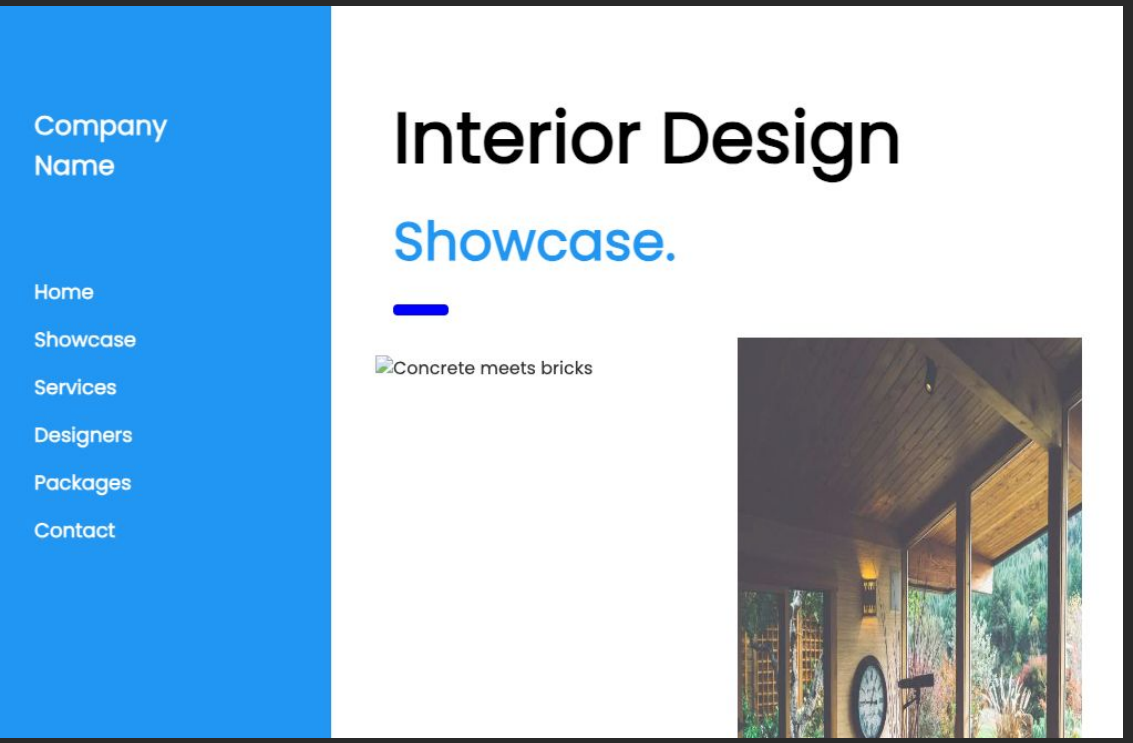
Imagine this situation: Evie and Grettel are working on this website together.

To make sure they're on the same page, they use GitHub to keep track of their files.

Source:
Website template from w3schools

Imaginary Scenario

Without Pull Requests



Evie decides the website would look better in a different colour. She changes the code to make it blue and updates GitHub.

- While she's changing the code, she makes a typo and it causes one of the images to disappear.
- She doesn't realise the mistake and uploads the changes to GitHub anyway.
- When Grettel goes to get the most updated version of the website, the error gets carried across!

Source:
Website template from w3schools

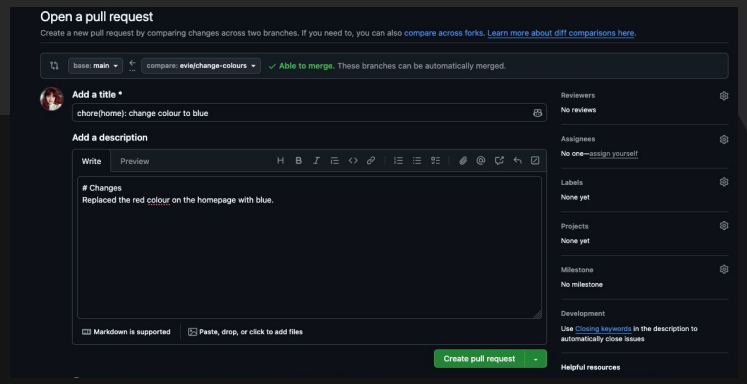
How could they have avoided this issue?

By using Pull Requests!

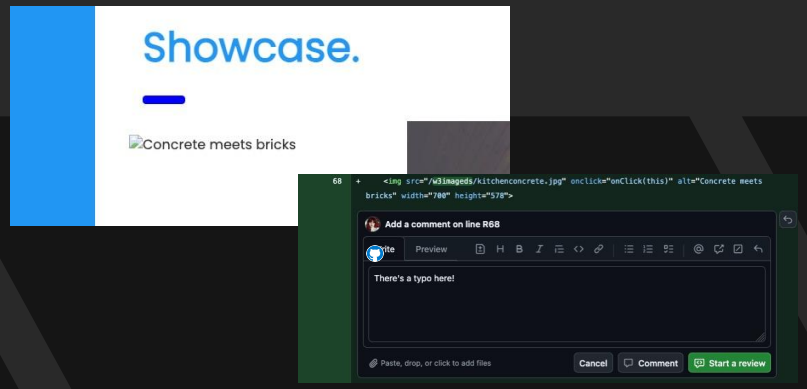


Imaginary Scenario

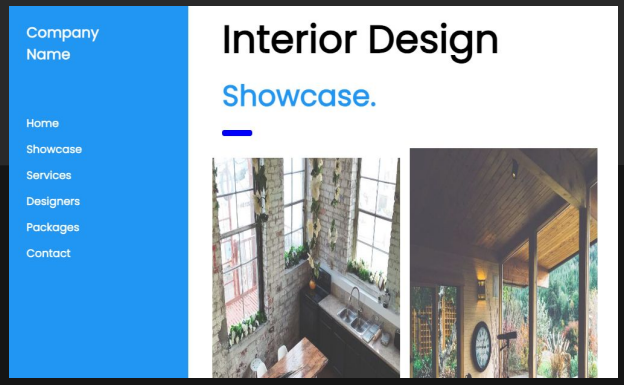
With Pull Requests



Instead of putting her code straight into the “main” part of their GitHub repository, Evie opens a ‘Pull Request’.



Grettel looks at the code and reviews the pull request. She notices the missing image and points it out to Evie.

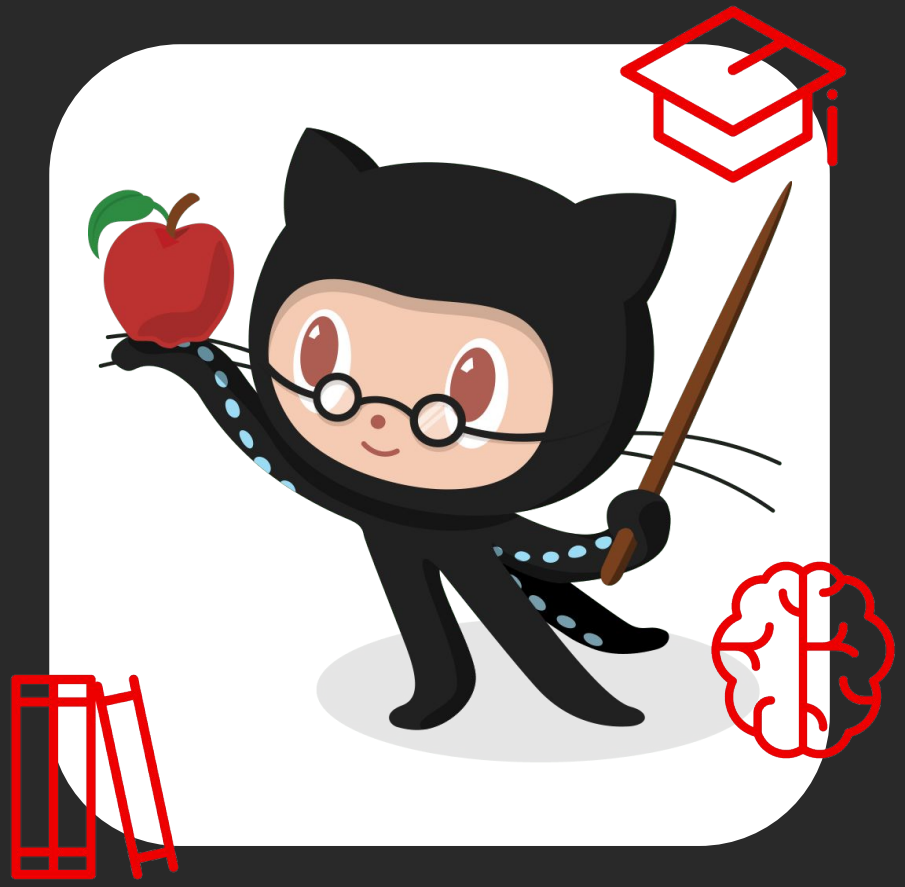


Evie fixes the error in her code, and Grettel checks the code again. It all looks good, so she approves it, and they add the code to the repository.

By using pull requests, we can check each others code and make sure that our code stays safe and free from mistakes!

Source:
Website template from w3schools
docs.github.com

Want to learn more about Pull Requests?



Additional Resources

"How to create a pull request in 4 min | GitHub for Beginners"

This video gives an overview of pull requests and shows you how to use them in ~4 minutes.

Review Pull Requests - Hands-On

Interactive GitHub Skills exercise: learn how to open, review, and merge pull requests in < 1 hour.

Source:
docs.github.com
github.com/skills

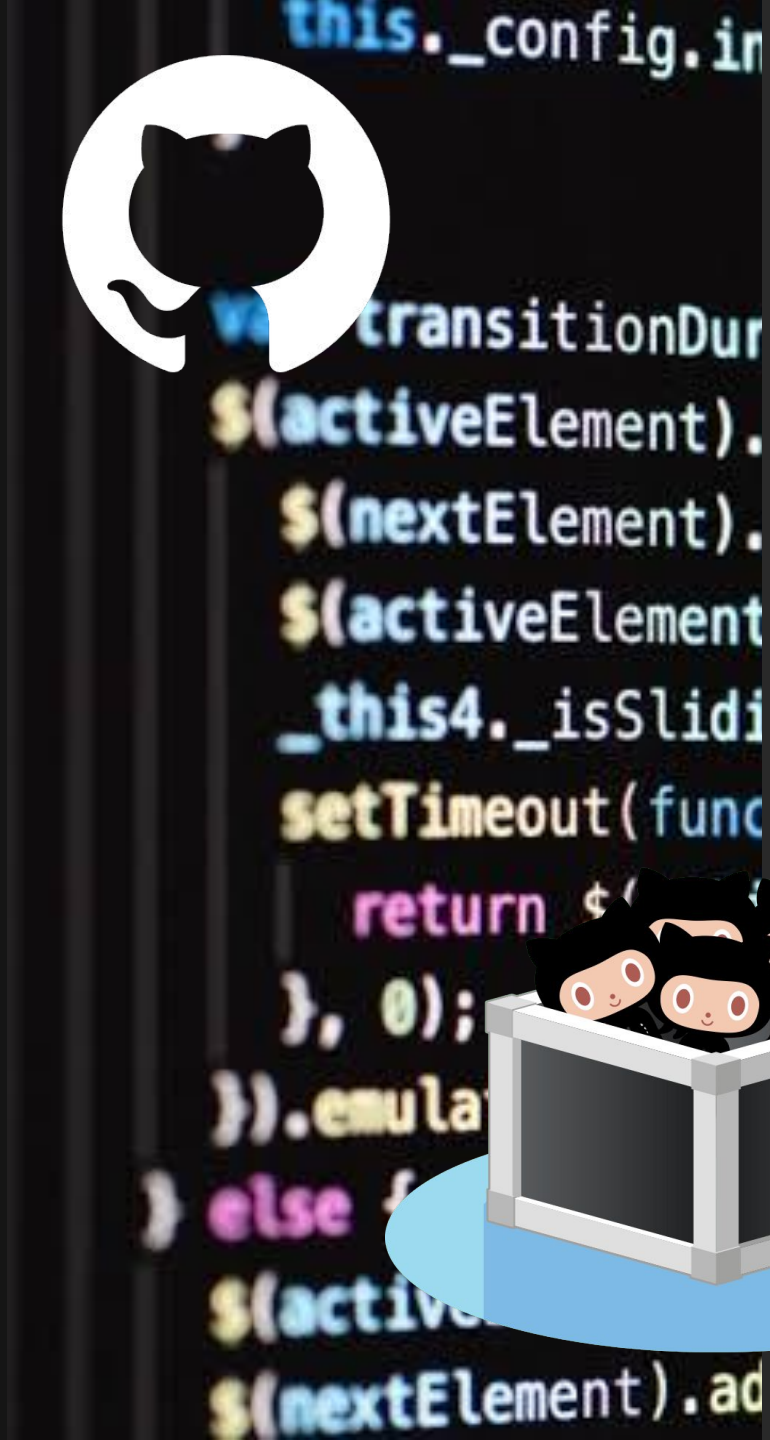
Hands-On Exercise

You will create a GitHub account, make your first pull request, and contribute a blog post to rh-ty-blog.github.io.

- https://rh-ty-blog.github.io/git-guide/
- https://github.com/rh-ty-blog/rh-ty-blog.github.io







Thank you

We hope you enjoyed! If you want to take the next step, you can try these:

- https://tutors.dev/course/git-site-test - comprehensive introduction to Version Control, suitable for beginners and intermediate level
- learn.github.com/skills - various hands-on exercises from complete beginner to advanced.
- github.com/skills/github-pages - step-by-step make your own GitHub pages site, just like the rh-ty-blog site.
- github.com/Santosh-Prasad-Verma/github-badges - collect some badges for your GitHub profile by completing simple tasks.



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